Khushaan Virk

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Education

University of British Columbia — Vancouver, BC

Bachelor of Applied Science (BASc) in Electrical Engineering — *Expected May 2028*

- **Relevant Coursework:** Circuit Analysis, Digital Logic Design, Signals & Systems, Embedded Systems Design

Experience

Software Engineer — Jan. 2026 – Present

Advizr — *Vancouver, BC*

- Built Python backend services integrating Claude and OpenAI APIs to automate lead generation, content workflows, and operational pipelines for 3 clients across real estate, fintech, and healthcare
- Shipped FastAPI backends and scheduled automation pipelines on Railway with Supabase as the data layer and Vercel-hosted client frontends, eliminating 30+ hours/week of manual workflows across deployed client systems
- Owned client engagements end to end from open-ended business goals to production, including a Canada-wide lead generation engine producing 11,000+ leads with on-demand rescraping

Sports Instructor / Summer Camp Leader — Sep. 2023 – Sep. 2025

City of Surrey — *Surrey, BC*

- Led structured programs for groups of 15–30, coordinating daily operations across staff and volunteers while maintaining safety standards
- Adapted instruction in real time to participants of varying skill levels, communicating clear expectations to children, parents, and co-instructors while resolving conflicts and incidents on the spot

Projects

65% Mechanical Keyboard PCB | *Altium, STM32F072, 4-Layer, Bare-metal C, USB HID* — Aug. 2026

- Designed a 67-key 4-layer keyboard PCB end to end in Altium around an on-board STM32F072CBT6 with crystal-less USB (HSI48 disciplined by CRS); derived the USB differential-pair geometry from the IPC-2141 microstrip approximation instead of a vendor calculator and routed it at 0.30/0.20mm for 91.0 Ω against a 90 Ω target, on the top layer over a solid ground plane with no vias
- Retuned Altium's stock solder-mask rule to JLCPCB's published 0.1mm capability, cutting 194 false DRC violations to 2 real ones and exposing a diode 0.152mm off its grid position that would have shipped as a dead key; the board returned from JLCPCB working on the first spin, 67 of 67 keys, no respin
- Brought the board up over SWD with OpenOCD and GDB before pressing a key, proving the 48 MHz clock by measuring SysTick at 2023 ticks in two seconds, and recovered the key matrix map from silicon by tweezer sweep rather than trusting the layout file (per-row counts 15/15/14/14/9)
- Ships bare-metal STM32F072 firmware with no HAL and no CubeMX, built on CMSIS startup and a hand-written Makefile: 1 kHz COL2ROW matrix scan, 5 ms deferred debounce, 8-byte boot-protocol report, USB HID via TinyUSB

Pedestrian Dead Reckoning (Phone IMU) | *Python, NumPy, SciPy, pandas, Matplotlib* — Jul. 2026

- Built a Python signal-processing pipeline in NumPy and SciPy to reconstruct a walking path from phone accelerometer and gyroscope logs alone, band-passing acceleration magnitude to the 0.7–2.1 Hz gait band and running peak detection for footfall counting, with cadence cross-checked against an FFT of the walk
- Estimated gyroscope bias from a stationary window, integrated the corrected rate for heading, and dead-reckoned an (x, y) trajectory step by step, measuring inertial drift by loop closure over a 76.2 m route at 5.1 to 6.0 m, roughly 7% of distance walked, with step length held out across two runs to avoid overfitting

Field-Following Autonomous Robot | *STM32, C, MOSFETs, I2C, IR* — Apr. 2026

- Built a dual-microcontroller robot with MOSFET H-bridge motor drivers and opto-isolated control lines for electromagnetic noise reduction during inductive sensor readings
- Implemented magnetic field detection using pick-up inductors to autonomously follow a guide wire, with a third sensor for intersection detection and pre-configured path navigation
- Designed an IR remote controller on a separate MCU transmitting 38kHz modulated commands for manual control and path selection
- Integrated VL53L0X Time-of-Flight laser ranging modules over I2C for real-time collision detection and avoidance in autonomous mode

Reflow Oven Controller | *Op-Amp, ADC, SSR, FSM, Python* — Feb. 2026

- Built a Python serial data acquisition framework around the firmware: real-time strip-chart plotting and automated Excel report generation over UART at 115200 baud, used as the verification harness during firmware tuning
- Wrote interrupt-driven embedded firmware implementing a 5-state FSM with PWM-based SSR control for precise thermal profile tracking through preheat, soak, reflow, and cooling phases
- Designed a closed-loop temperature control system using a K-type thermocouple with cold-junction compensation (LM335) and OP07 op-amp signal conditioning, achieving ±3°C accuracy over 25–240°C

Technical Skills

Embedded & Control Systems: ARM (STM32F072), bare-metal C, USB HID, SWD bring-up, ADC, I2C, SPI, UART, PWM, FPGA (Nios V soft core)

Programming: Python, C, Embedded C, SystemVerilog, SQL

PCB Design: Altium Designer, KiCad, EasyEDA, 4-layer stackup, controlled-impedance routing, DRC/DFM, JLCPCB fab and assembly

Instrumentation & Tools: Git, OpenOCD/GDB, ST-Link, oscilloscopes, logic analyzers, SMD soldering, signal conditioning, MATLAB, PuTTY